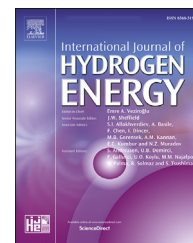


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Performance evaluation of a combined power generation system integrated with thermochemical exhaust heat recuperation based on steam methane reforming

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HIGHLIGHTS

- Concept of a combined cycle power plant with chemically recuperated gas turbine.
- Effect of pressure, temperature and steam-to-methane ratio on efficiency is studied.
- Thermochemical recuperation leads to an increase in efficiency on 4–7%.
- Maximum efficiency is observed at $T_{in}=1600^{\circ}\text{C}$ in the pressure range of 20–25 bar.

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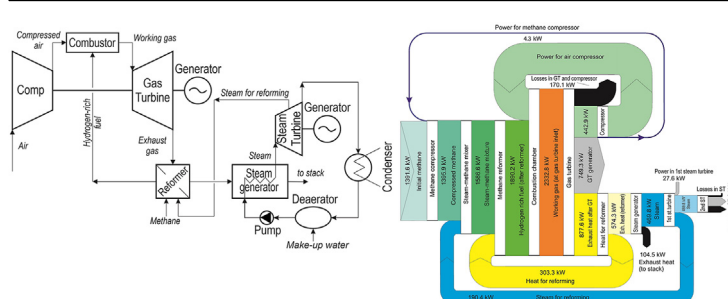
Gas turbine

Thermochemical recuperation

Hydrogen-rich fuel

Thermodynamic analysis

GRAPHICAL ABSTRACT



ABSTRACT

The present paper applies the thermodynamic analysis with the determining the efficiency of a combined cycle power plant with a chemically recuperated gas turbine. Thermochemical recuperation of exhaust heat after a gas turbine is realized via the steam methane reforming process. The main concept of combined cycle power plant (CCPP) with chemically recuperated gas turbine (CRGT) is based on the use of exhaust heat for endothermic reforming of the original hydrocarbon fuel in a reformer and for steam generation for a steam cycle. To understand the effect of operating variables such as temperature, pressure, and steam-to-methane ratio on the overall efficiency, the energy and mass balances were compiled. The energy flows were represented by a Sankey diagram. The results of the thermodynamic analysis show that efficiency of CCPP with CRGT is significantly higher (4–7%) than efficiency of CCPP with a conventional gas turbine without TCR. Maximum efficiency of CCPP with CRGT of 0.6412 is observed at inlet temperature of working gas of 1600°C , pressure of 23 bar for a steam-to-methane ratio of 3.0. In the temperature of inlet working gas below 1200°C the increase in the efficiency of CCPP with TCR is less than 2%.

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Introduction

The challenge to reduce carbon dioxide emissions is one of the key ones in world science and industry at present. Global warming is inextricably linked with the emission of carbon dioxide [1]. In this regard, various international agreements have been adopted to reduce CO₂ emissions [2,3]. The energy industry is one of the main sources of CO₂ emissions in the world [4–6]. Despite of advances in renewable energy, the share of fossil fuels in the energy balance are remaining significant. A rapid refusal of fossil fuel consumption is associated with great economic disruption; theretofore, a smooth transition to a low-carbon industry can take several decades. In this regard, one of the main ways to reduce CO₂ emissions is to increase energy efficiency [7]. To reduce CO₂ emissions in the existing industry, we must increase its efficiency.

One of the important consumers of fossil fuels, in particular natural gas, are gas turbine units (GTU) designed to generate electrical energy [8,9]. According to the first law of thermodynamics, one of the ways to increase the energy efficiency of such units is to increase the temperature of the working gases at the gas turbine inlet. Thus, modern samples of gas turbines have a temperature of combustion products at turbine inlet above 1500–1600 °C [10]. An increase in the temperature of the combustion products leads to an increase in the temperature of the flue gases after a gas turbine. Therefore, an important direction in increasing the efficiency of GTUs is the beneficial use of exhaust heat after the gas turbines.

The addition of a steam power cycle to the gas turbine cycle is a widespread practice to increase the efficiency of gas turbine units [11–13]. The exhaust heat after the gas turbine is used to produce steam, which is then used in a steam turbine. This principle is called combined, and installations with a gas turbine and a steam turbine are called combined cycle plants. As a rule, steam for a steam turbine has a temperature of 400–500 °C at a pressure of up to 40–50 bar [14–16]. This ensures the efficiency up to 60–65% [17,18]. Many schemes of combined cycle plants have been proposed: classic, steam injection [19,20], supplementary fuel firing [21], etc.

In addition, one of the ways to improve the efficiency of gas turbines is the use of thermochemical exhaust heat recuperation [22,23]. Gas turbine units with thermochemical recuperation are called chemically recuperated gas turbines (CRGT) [24–26]. The main operating principle of CRGT is the use of exhaust heat for endothermic reforming of the original hydrocarbon fuel. In such turbines, the products of fuel reforming (synthesis gas) are further used as fuel in the combustion chamber instead of the initial fuel. In the high-temperature CRGTs, a steam reforming reaction is favorable for the endothermic transformation of natural gas (methane) [27]. In this case, the exhaust heat after a gas turbine is used for steam generation (to occur methane reforming) in a steam generator and for the endothermic methane steam reforming process in a methane reformer.

Olmsted and Grimes firstly suggested to use an endothermic reaction of methane steam reforming in a gas turbine [28]. Authors proposed the addition of a steam methane reformer after the gas turbine to produce synthesis gas and

further use of the reforming products as a fuel in a combustion chamber. Olmsted and Grimes presented that a steam-methane mixture absorbs exhaust heat thermally (as the temperature of the mixture increases), and chemically (as the endothermic process occurs), resulting in a higher potential recuperation of exhaust heat than can be observed by conventional recuperation using only air preheating.

The different diagrams of the chemically recuperated gas turbines were investigated in the last years [25,29]. Sadeghi et al. [30] performed a comparison of the organic Rankine cycle system with CRGT based on steam methane reforming. The authors showed that CRGT had a lower sum of the unit cost of product value, which indicates a better economic performance. Carapelucci and Milazzo [24] investigated a reheat gas turbine with thermochemical recuperation of exhaust heat based on the steam methane reforming process. Authors showed that efficiency of CRGT increases by 14.1% compared to a conventional design. Su et al. [31] proposed a biogas-fired cogeneration system based on a chemically recuperated gas turbine cycle. The authors suggested to use a reformer to endothermically transform of bio-gas and then to burn the reforming products in a combustion chamber.

A concept of chemically recuperated gas turbines can be considered as an on-board hydrogen production technology [32,33]. A hydrogen-rich fuel (synthesis gas with a high molar fraction of H₂) is produced in GTU before using in the same GTU [34,35].

The development of CRGT technology requires the search for new engineering solutions to improve the efficiency of such turbines. Combining the CRGT with a steam power plant is a promising direction. One of the first works on the study of a combined cycle plant, in which exhaust heat after a gas turbine is used for steam methane reforming, is the article written by Verkhivker and Kravchenko [36]. Authors proposed to use exhaust heat for steam methane reforming. In parallel, they proposed to use steam for the reforming reaction from a steam turbine. Authors showed the high efficiency of such a solution. However, in their scheme, synthesis gas was supplied for external use.

Previously, the effect of operation parameters on the efficiency of single chemically recuperated gas turbines fired with methane [37] and ammonia [38] has been investigated by author of the present paper. The results of investigation of CRGT fired with methane showed that temperature, pressure, and steam-to-methane ratio have a significant effect on the turbine efficiency. However, in the investigated schematic diagrams of a chemically recuperated gas turbine, steam for reforming was produced in a steam generator and supplied directly to a reformer after a steam generator. The main difference of the present paper is investigation of a combined cycle power plant with a chemically recuperated gas turbine. However, there are no published results on the efficiency of the combined cycle power plants with the chemically recuperated gas turbines as a function of operating parameters.

The main goal of this article is to study a combined cycle plant with thermochemical exhaust heat recuperation, in which the synthesis gas after the reformer is used as a fuel. In such plant, steam for reforming is extracted from a steam turbine. The efficiency of a chemically recuperated gas turbine essentially depends on the parameters of the working gases at

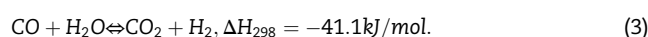
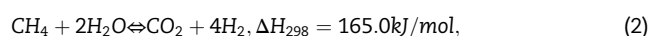
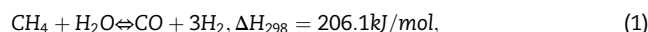
the turbine inlet as well as on operating parameters such as temperature, pressure, and steam-to-methane ration of an inlet reaction mixture. On the other hand, the steam extraction from a steam turbine for methane reforming will affect the efficiency of a steam turbine cycle. Therefore, in this article, the effect of operating parameters on the thermodynamic efficiency of a combined cycle plant with thermochemical exhaust heat recuperation is established. In such plant, exhaust heat after a gas turbine is used for methane reforming and steam generation for a steam cycle; steam for the steam methane reforming process is supplied from a steam turbine; synthesis gas after a reformer is used as fuel in a combustion chamber.

Combined cycle with CRGT concept

The main concept of a combined cycle power plant (CCPP) with a chemically recuperated gas turbine (CRGT) is based on the use of exhaust heat for endothermic reforming of the original hydrocarbon fuel in a reformer and for steam generation for a steam cycle. Natural gas is one of the main fossil fuels for power plants. Moreover, as shown in the author's previous work, the thermochemical exhaust heat recuperation system based on steam methane reforming has the maximum efficiency for high-temperature exhaust gases [27]. Fig. 1 shows a schematic diagram of a combined cycle power plant (CCPP) with a chemically recuperated gas turbine (CRGT) based on steam methane reforming.

According to the schematic diagram in Fig. 1, exhaust heat after a gas turbine is sequentially used in a reformer and steam generator. The efficiency of the steam methane reforming process and methane conversion is strongly dependent on the temperature. Therefore, high-temperature exhaust gas is favorable for steam methane reforming. On that score, a reformer is installed before a steam generator. Partially cooled exhaust gases after the reformer are used in the steam generator to generate steam for a steam cycle.

In a reformer, the heat source for the endothermic steam methane reforming process is exhaust heat after a gas turbine. In parallel, steam and methane are supplied to a reaction space of a reformer where the steam methane reforming process occurs over the catalysts (for example, industrial Ni-based catalyst [39,40]). The steam methane reforming process includes various chemical reactions [41,42]. However, according to a thermodynamic analysis of the previous investigations [42,43], only the following three reactions have a significant effect on heat and mass balance in steam methane reforming in the temperature range of 500–1200 °C are negligible:



Steam for the methane reforming process is supplied from a steam turbine (steam extraction). According to Le Chatelier's principle, pressure also has a significant effect on a methane conversion. Thus, regulation steam pressure for reforming is an important task. Residual steam classically works in a steam cycle. To compensate the steam extraction for a reforming, make-up water is added to cycle in a deaerator.

The schematic diagram in Fig. 1 has ample scope for regulation. Operational parameters such as temperature, pressure, and composition of the inlet steam-methane mixture have a significant effect on the efficiency of CRGT. The temperature of the exhaust gases is controlled by the expansion rate in the turbine. The steam pressure and mass flow rate are controlled by steam extraction in a steam turbine. It should be noted that the steam-methane ratio also affects methane conversion. However, an increase in steam extraction from a steam turbine for reforming leads to a reduction in power generation in a steam turbine. To establish the relationship between the various operating modes of a

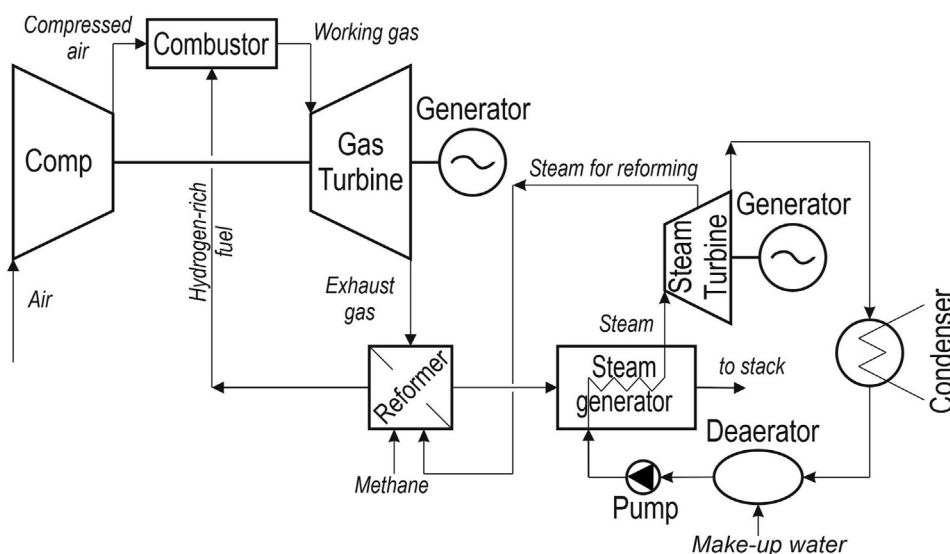


Fig. 1 – A schematic diagram of a combined cycle power plant (CCPP) with a chemically recuperated gas turbine (CRGT) based on steam methane reforming.

combined cycle power plant (CCPP) with a chemically recuperated gas turbine (CRGT) allows thermodynamic analysis.

Hydrogen-rich fuel after a reformer is supplied to a combustion chamber of a gas turbine. Therefore, initial methane is replaced by new synthetic fuel with a high molar fraction of hydrogen. Hydrogen-rich gas as a fuel in the combined cycle power plant with the chemically recuperated gas turbine is also one of the important advantages of such a concept. There are sufficient results in the scientific articles that replacing an original hydrocarbon fuel with the products of its reforming leads to decreasing in NO_x emission [44,45]. Lui et al. studied the combustion characteristics in a gas turbine fired with methane and the products of its reforming [46]. Authors showed that thermochemical recuperation can be considered as an ultralow NO_x emission technology.

Thermodynamic analysis

Cycle analysis

The thermodynamic analysis of the combined cycle power plant with chemically recuperated gas turbine has been performed with the help of Aspen HYSYS [47–49]. This software is a widely used process simulator for the modeling of the different schemes including the schemes with chemical reactions. Aspen HYSYS makes it possible to model compression and expansion, combustion, chemical reaction, etc. Based on the developed model, the thermodynamic analysis was conducted for a wide range of operating parameters. This analysis showed the overall efficiency of CCPP with CRGT as a function of operating parameters.

A computational scheme of the concept of CCPP with CRGT from Fig. 1 in Aspen HYSYS is shown in Fig. 2.

In the scheme from Fig. 2, exhaust heat (6) is supplied firstly to a reformer and then to a steam generator. A steam turbine is virtually divided into two parts: steam turbine1 and steam turbine2. Steam (13) after a steam generator is supplied to a steam turbine1. After a steam turbine1, the flow of steam is divided into two streams: one stream (10) is fed for reforming, and another stream (14) is fed to a steam turbine2. In a reformer, the steam methane reforming process is occurring. In a combustor, the combustion of hydrogen-rich gas (4) is taking place. Temperature of working gas (5) before a gas turbine is regulated by the air mass flow rate (9) with the help of a useful function (Adjust). This function makes it possible to set the temperature of the working gas as the target variable and an air mass flow rate as an adjusted variable.

In the scheme from Fig. 2, the heat flow of exhaust gas (6) is divided into three terms: heat for reforming, heat for steam generation, and heat with exhaust gas to stack. A heat balance equation has the following form:

$$H_{\text{exh}} = H_{\text{ref}} + H_{\text{st.gen}} + H_{\text{stack}} \quad (4)$$

where H_{ref} , $H_{\text{st.gen}}$, H_{stack} is enthalpy of exhaust gas supplied in a reformer, steam generator, respectively, and residual heat passed to stack.

A net work output in the scheme from Fig. 2 has been determined as a sum of all input and output flows based on the following expression:

$$W_{\text{CCPP}} = W_{\text{gt}} + W_{\text{st1}} + W_{\text{st2}} - W_{\text{a.comp}} - W_{\text{m.comp}} - W_{\text{pump}} \quad (5)$$

where W_{gt} , W_{st1} , W_{st2} is the power produced in a gas turbine, steam turbine1, steam turbine2, respectively; $W_{\text{a.comp}}$, $W_{\text{m.comp}}$ is the power for air and methane compressing, respectively; W_{pump} is the power consumed in a pump.

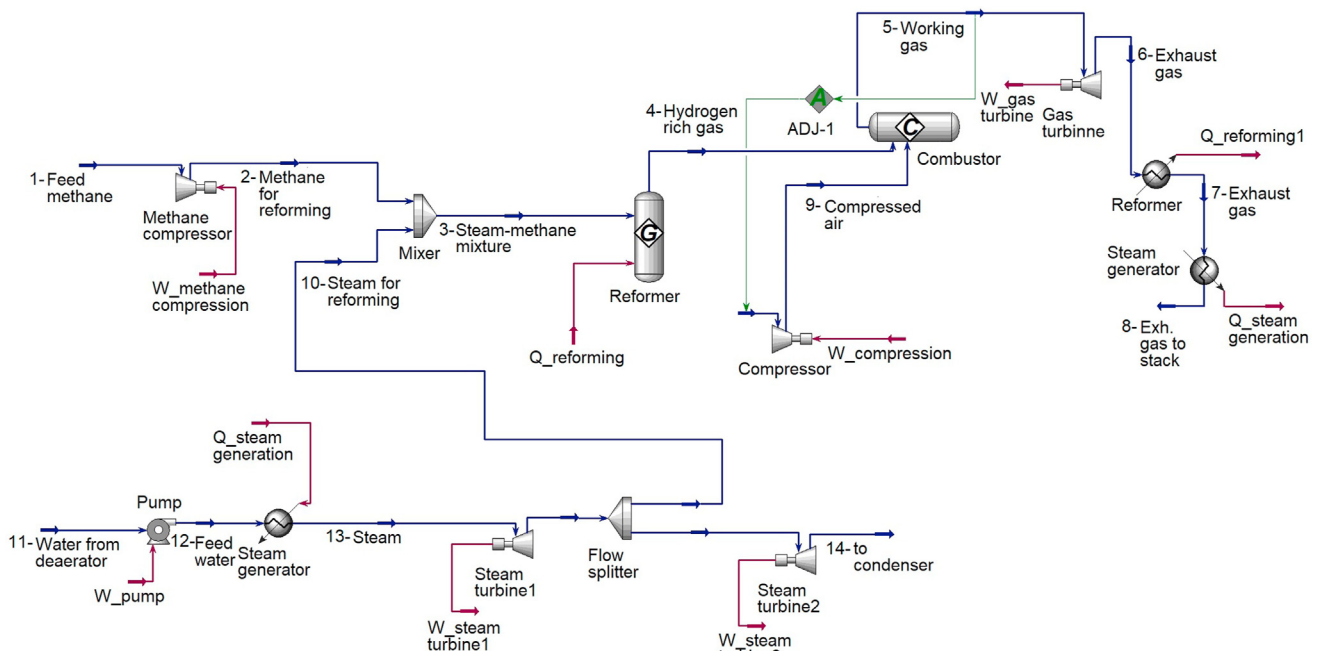


Fig. 2 – A computational scheme of a combined cycle power plant with a chemically recuperated gas turbine in Aspen HYSYS.

A scheme of CCPP without thermochemical recuperation when whole steam after a steam generator is fed to a steam turbine was also considered in the thermodynamic analysis. No steam extraction for reforming from a steam turbine. In this scheme, methane was supplied directly to a combustor. This scheme was analyzed to compare the efficiency of CCPP with CRGT and CCPP with a conventional gas turbine.

The efficiency for various configurations of CCPP with CRGT was determined to understand the effect of operational parameters. The efficiency of CCPP with CRGT was calculated as a following [50,51]:

$$\eta_{CCPP} = \frac{W_{CCPP}}{\dot{m}_{CH_4} \cdot LHV_{CH_4}} \quad (6)$$

where W_{CCPP} is a net work output of CCPP with CRGT; $\dot{m}_{CH_4}$ is the mass flow rate of methane; LHV_{CH_4} is the methane lower heating value.

Model setup

The thermodynamic analysis was performed under a wide range of operational parameters. The initial conditions and turbine parameters for a thermodynamic analysis are presented in Table 1. Operational parameters of CCPP with CRGT were mainly varied in a CRGT part because temperature, pressure, and inlet steam-to-methane ratio significantly affect the efficiency of such turbines. Operational parameters of a steam cycle part were constant for all cases: pressure of 40 bar and temperature of inlet steam of 400 °C. Such parameters are widely used in the real CCPP. A change of pressure before a steam turbine will have no significant effect on changing the overall efficiency of CCPP with CRGT in comparison to CCPP with a conventional gas turbine. Therefore, pressure of 40 bar was set for all cases. Simultaneously, pressures for reforming were varied notably from 10 to 40 bar with a step of 5 bar. Also, it was assumed that temperature of exhaust gas after a steam generator is 100 °C for all cases.

Table 1 – Stream properties and efficiency parameters of the gas and steam turbines, the air and methane compressors.

Parameter	Value
CH4 mass flow rate	100 kg/h
Gas temperature at GT inlet (T _{in})	1200–1600 °C (with step of 100 °C)
Compression/expansion ratio	10–40 (with step of 5)
Pressure losses	0
Turbine isentropic efficiency	90%
Compressor isentropic efficiency	85%
Steam temperature	400 °C
Steam pressure	40 bar
Condenser pressure	0.06 bar
Steam-to-methane ratio (mol-to-mol)	1.0; 2.0; 3.0
Temperature of atmospheric air and methane	20 °C
Pressure of all inlet streams (except air)	10 bar

SMR analysis

A share of exhaust heat used in a reformer is depending on methane reforming enthalpy. In parallel, the enthalpy of methane reforming with steam is depending on process variables such as temperature, pressure, and inlet steam-to-methane ratio. Moreover, these variables are also affecting on reforming product distributions after a reformer.

The equilibrium conditions of the steam methane reforming system are calculated by using the Gibbs free energy minimization method with using the Peng-Robinson equation of state. The detailed algorithm and governing equations were presented in author's previous papers [37,38]. The thermodynamic analyses of steam methane reforming under various operating variables were conducted via Aspen HYSYS software.

The thermodynamic analysis makes it possible to calculate the equilibrium concentration of the reforming products at the given operating parameters such as temperature, inlet steam-to-methane mixture and pressure. The main concept of the Gibbs free energy minimization method is the fact, that the chemical system is thermodynamically favorable when its total Gibbs free energy is minimum and its differential is equal to zero [52,53]:

$$dG^t = 0 \quad (7)$$

where G^t is total Gibbs free energy.

Total Gibbs free energy of the SMR decomposition process can be calculated as a sum of the chemical potential of all components:

$$G^t = \sum n_i \mu_i \quad (8)$$

where n_i is number of i th component moles; μ_i is chemical potential of i th component.

A chemical potential of each component can be calculated based on following expression:

$$\mu_i = \Delta G_{fi}^0 + RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (9)$$

where G_{fi}^0 is standard Gibbs free energy of i th component formation; R is molar gas constant; T is system temperature; f_i is fugacity of a pure i th component; f_i^0 is fugacity of a pure i th component at the standard state.

If the equations presented above will be combined, the equation for total Gibbs free energy of the chemical system can be written as the following:

$$G^t = \sum n_i \Delta G_{fi}^0 + \sum n_i RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (10)$$

The methane conversion is one of the main criteria of efficiency of the steam methane reforming process. The equilibrium methane conversions are determined as follows:

$$X_{CH_4} = \frac{CH_{4,in} - CH_{4,out}}{CH_{4,in}} \quad (11)$$

where $CH_{4,in}$, $CH_{4,out}$ is the mole fraction of methane at a reformer inlet and outlet, respectively.

Also for further analysis, parameter β is introduced which is showing a ratio of steam to methane at a reformer inlet:

$$\beta = \frac{H_2O_{in}}{CH_{4,in}} \quad (12)$$

As mentioned above, one of the main ideas of thermochemical exhaust heat recuperation is using exhaust heat for an endothermic transformation of initial hydrocarbon fuel. Such transformation leads to an increase in combustion heat of new hydrogen-rich fuel in comparison with combustion heat of initial fuel (in the calculation for 1 kg of initial fuel).

A ratio of a lower heating value (LHV) of new hydrogen-rich fuel produced by reforming 1 kg of initial methane in a reformer to a lower heating value (LHV) of 1 kg of initial methane is named a heat transformation coefficient (HTC) [27]. According to definition, heat transformation coefficient is defined as follows:

$$n_q = LHV_{hyd,rich} / LHV_{CH_4} \quad (13)$$

where LHV_{syngas} is a lower heating value of hydrogen-rich gas produced by reforming of 1 kg of initial methane, kJ; LHV_{fuel} is a lower heating value of 1 kg of initial methane, kJ.

Results and discussion

Reformate composition and enthalpy

The thermodynamic analysis of steam methane reforming was carried out for the different operating parameters. A comparison of the results of synthesis gas composition obtained in this paper with the results of other authors is presented in Table 2. The upper results are calculated via Aspen HYSYS, the middle rows are the mole fraction from the paper presented by Verkhivker and Kravchenko [36], the bottom results are the mole fraction of each component that was calculated based on the classical thermodynamic analysis [54]. A good correlation is observed for all results.

The equilibrium composition of the products of steam methane reforming is depending on temperature, pressure, and steam-to-methane ratio according to Le-Chatelier's principles. An increase in temperature and steam-to-methane ratio leads to an increase in methane conversion. On the

other hand, an increase in pressure leads to a decrease in methane conversion. Table 3 exhibits the equilibrium composition of the reforming products at different operation parameters: temperature, pressure, and steam-to-methane ratio.

An increase in temperature leads to an increase in the methane conversion and the hydrogen yield in the synthesis gas. This is because the change in the Gibbs energy of a reforming reaction decreases as the temperature increases. A steam addition has a significant effect on methane conversions and synthesis gas composition. A steam addition in the inlet steam-methane mixture is increasing the methane conversion; especially in the temperature range below 800 °C. In the temperature range above 800 °C, the effect of steam addition on the methane conversion is decreasing because the temperature level is enough to get almost full conversion of methane. This fact is in good agreement with the works of Ozkara-Aydinoglu [53] and Hegarty et al. [55].

It should be noted that steam addition in the feed stream to a reformer for the combined cycle power plant (CCPP) with a chemically recuperated gas turbine (CRGT) leads to a decrease in power output in a steam turbine. Moreover, the use of high-pressure steam for reforming also reduces the power output of the steam turbine because in this case methane conversion is minimal.

Table 3 also shows the effect of pressure on the reformate composition. An increase in pressure reduces the methane conversion; especially at temperatures below 800 °C. At temperatures above 800 °C, the effect of pressure on the methane conversion is decreasing.

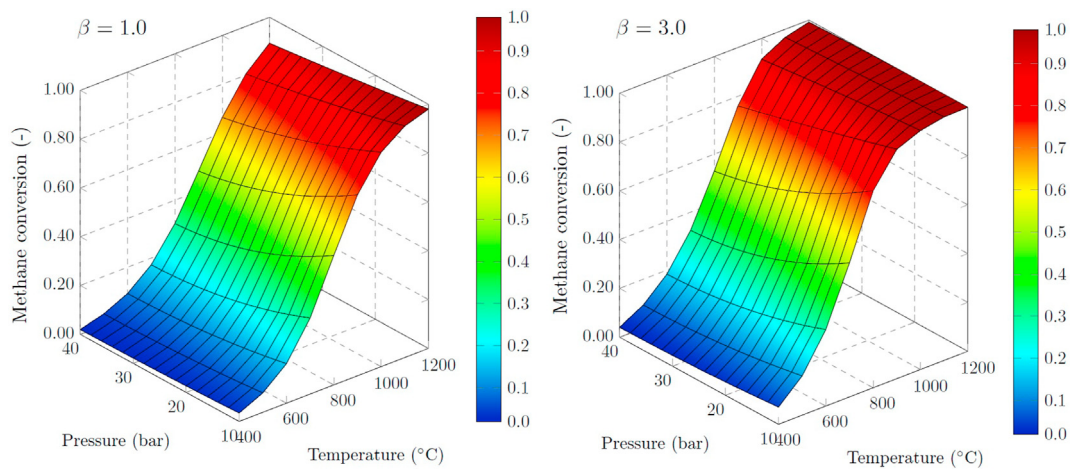
Le-Chatelier's principles are well illustrated by the effect of temperature, pressure, and inlet steam-to-methane mixture on methane conversion. Fig. 3 depicts the equilibrium methane conversion as a function of temperature and pressure when the different steam-to-methane ratios are fed to a reformer. It is observed that higher steam amounts in feedstock resulted in more methane conversion, especially at moderately low pressures. However, an increase in pressure leads to a decrease in methane conversion because the number of moles of the reforming products is higher than the number of moles of initial reactants. Therefore, according to Le-Chatelier's principle, the system will oppose changes in the system to achieve a new equilibrium state.

Table 2 – The comparison of the synthesis gas composition at a steam-to-methane ratio of 2, % (mol/mol).

Temperature	Pressure	Source	Gas composition, %				
			H ₂	CO	H ₂ O	CO ₂	CH ₄
1000 K	1 bar	HYSYS	63.55	14.22	16.11	5.22	0.9
		Verkhivker [36]	63.42	14.43	16.11	5.12	0.92
		Classical [54]	63.01	14.32	16.32	5.23	1.12
1000 K	5 bar	HYSYS	61.08	14.85	17.57	5.13	1.55
		Verkhivker [36]	61.44	14.93	17.97	4.18	1.48
		Classical [54]	60.66	14.82	17.32	5.55	1.65
1100 K	1 bar	HYSYS	63.78	15.95	16.2	3.98	0.09
		Verkhivker [36]	63.76	16.11	16.11	3.94	0.08
		Classical [54]	63.88	16.08	16.02	3.93	0.09
1100 K	5 bar	HYSYS	61.24	14.71	18.11	4.28	1.66
		Verkhivker [36]	61.45	14.93	17.96	4.18	1.48
		Classical [54]	61.28	15.02	17.87	4.24	1.59

Table 3 – The equilibrium compositions of the reforming products at different operation parameters.

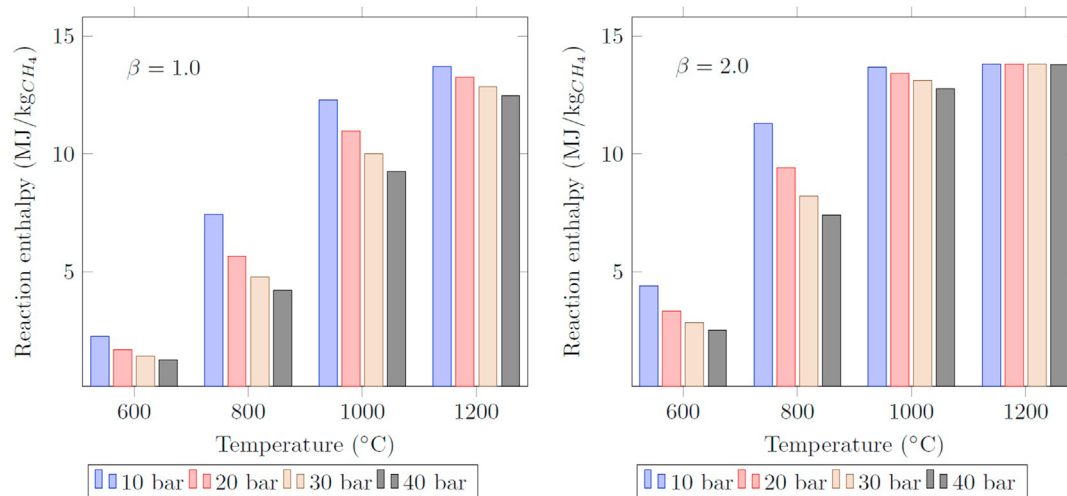
T, °C	β	p = 10 bar						p = 30 bar					
		H ₂	CO	CH ₄	H ₂ O	CO ₂	η_q	H ₂	CO	CH ₄	H ₂ O	CO ₂	η_q
400	1.0	0.0819	0.0003	0.4585	0.4391	0.0202	1.010	0.0531	0.0001	0.4729	0.4606	0.0132	1.006
	2.0	0.0847	0.0002	0.2976	0.5965	0.0210	1.015	0.0547	0.0001	0.3101	0.6215	0.0136	1.010
	3.0	0.0836	0.0002	0.2182	0.6772	0.0207	1.020	0.0539	0.0001	0.2294	0.7032	0.0134	1.013
600	1.0	0.2912	0.0211	0.3435	0.2873	0.0570	1.045	0.2032	0.0094	0.3933	0.3503	0.0438	1.029
	2.0	0.3002	0.0164	0.2011	0.4196	0.0627	1.068	0.2095	0.0074	0.2426	0.4936	0.0469	1.044
	3.0	0.2940	0.0138	0.1343	0.4948	0.0632	1.088	0.2059	0.0063	0.1701	0.5709	0.0468	1.057
800	1.0	0.5598	0.1435	0.1480	0.1163	0.0323	1.148	0.4370	0.0887	0.2368	0.1948	0.0427	1.096
	2.0	0.5490	0.1143	0.0567	0.2285	0.0515	1.200	0.4381	0.0719	0.1205	0.3139	0.0556	1.135
	3.0	0.5087	0.0894	0.0255	0.3163	0.0601	1.226	0.4189	0.0596	0.0703	0.3912	0.0600	1.164
1000	1.0	0.7029	0.2267	0.0349	0.0297	0.0057	1.245	0.6367	0.1949	0.0839	0.0715	0.0130	1.200
	2.0	0.6226	0.1695	0.0031	0.1764	0.0285	1.273	0.5946	0.1569	0.0200	0.1976	0.0309	1.250
	3.0	0.5385	0.1252	0.0009	0.2947	0.0407	1.273	0.5269	0.1206	0.0069	0.3043	0.0413	1.262

**Fig. 3 – Methane conversion as a function of operating parameters.**

Methane conversion is related to the enthalpy of the steam methane reforming process. In turn, the enthalpy determines an exhaust heat recuperated in a reformer. Fig. 4 shows the equilibrium changes in the enthalpy of the steam methane

reforming process for various operating variables such as temperature, pressure, and inlet steam-to-methane ratio.

It is observed that the maximum enthalpy takes place at temperatures above 1000 °C and at beta above 1. However, in

**Fig. 4 – Enthalpy of the steam methane reforming process as a function of operating parameters.**

the temperature range below 600 °C or above 1200 °C, the pressure effect is almost negligible.

In the temperature range of 800–1000 °C, the enthalpy significantly depends on the pressure and the share of steam in the inlet reaction mixture. That is why the thermodynamic analysis of the combined cycle power plant (CCPP) with a chemically recuperated gas turbine (CRGT) is an important task because the analysis will make it possible to establish the effect of these parameters on the efficiency of the entire power plant.

Cycle analysis

Operating variables such as temperature, pressure, and steam-to-methane ratio have a noteworthy effect on CCPP efficiency. Temperature of steam of 400 °C and pressure of 40 bar in a steam cycle of CCPP is stable for all cases. The mass flow rate of steam in a steam turbine is changing due to a steam extraction for the reforming process. In parallel, the operational parameters for a gas turbine cycle are significantly varied. Temperature of working fluid at a gas turbine inlet can be regulated by air flow rate. An increase in pressure in a gas turbine leads to an increase in a compressor power for methane compression. In addition, an increase in pressure and inlet steam-to-methane (β) ratio in a reformer leads to a decrease in the power of the steam turbine because steam extraction reduces the mass flow of steam through a steam turbine. In addition, a decrease in temperature of the working gases before the gas turbine leads to an increase in compressor power, which can lead to a decrease in the efficiency of the gas turbine cycle. Therefore, the efficiency of CCPP with CRGT depends on many variables. Fig. 5a shows the dependence of efficiency of the combined cycle power plant (CCPP) with the chemically recuperated gas turbine (CRGT) as a function of operating parameters such as temperature, pressure, and inlet steam-to-methane ratio.

Fig. 5a shows that an increase in temperature leads to an increase in efficiency CCPP for all cases. This fact is

corresponding to the second law of thermodynamics. However, the effect of pressure and steam-to-methane ratio on the efficiency is significant.

When the temperature of the working gases at the gas turbine inlet is 1200 °C, an increase in the CCPP efficiency with an increase in a steam-to-methane ratio is minimal, especially in the pressure range above 25 bar. This is due to the fact that in this case the enthalpy of the process is minimal and the increase in a steam-to-methane ratio does not have a noticeable effect on recuperated heat in a reformer. In addition, an increase in steam extraction for reforming leads to a decrease in the power of the steam turbine. Therefore, it is clear that an increase in a steam-to-methane ratio for the low-temperature gas turbines does not significantly increase the efficiency of the CCPP with CRGT.

When the working gas temperature of 1400 °C, the effect of pressure and steam-to-methane ratio on the efficiency is more noticeable. An increase in pressure leads to an increase in the efficiency of the CCPP. However, the maximum increase in efficiency occurs at pressures below 25 bar. This is also related to the enthalpy of the process. At pressures below 25 bar, the enthalpy of the reforming process is highly dependent on a steam-to-methane ratio. Fig. 5a depicts that increasing a steam-to-methane ratio for CRGT with a high expansion ratio does not lead to a significant increase in the efficiency of the CCPP.

The most interesting case is the effect of pressure and a steam-to-methane ratio at a gas temperature of 1600 °C. An increase in a steam-to-methane ratio leads to a marked increase in the CCPP efficiency. When $\beta = 1$ an increase in pressure leads to a monotonous increase in the CCPP efficiency at pressures above 15 bar. When $\beta = 2$ an increase in pressure above 25 bar has no effect on efficiency. This is due to the fact that although the increase in pressure leads to an increase in the gas turbine power, it reduces the share of recuperated heat in the reformer because the enthalpy of the process decreases. For $\beta = 3$ the effect of pressure on efficiency is most pronounced. In the pressure range up to 20 bar, the

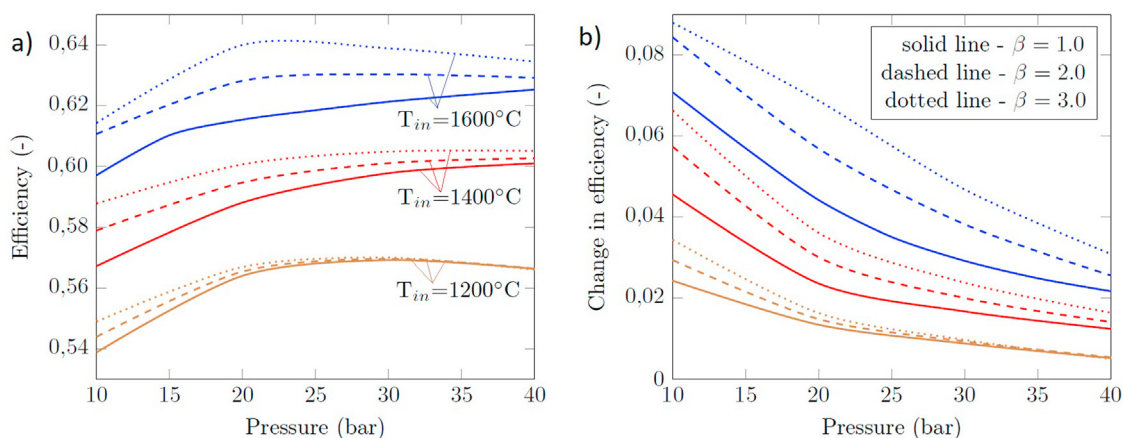


Fig. 5 – a - Efficiency of the combined cycle power plant (CCPP) with a chemically recuperated gas turbine (CRGT) as a function of operating parameters such as temperature, pressure, and inlet steam-to-methane ratio; b - Change in the efficiency of CCPP with CRGT in comparison with CCPP without CRGT: solid lines – $\beta = 1.0$; dashed lines – $\beta = 2.0$; dotted lines – $\beta = 3.0$; blue lines – $T_{in} = 1600$ °C; red lines – $T_{in} = 1400$ °C; brown lines – $T_{in} = 1200$ °C. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

efficiency increases monotonically. Further, at $p = 23$ bar, it reaches the maximum level and, as the pressure increases above 23 bar, it monotonically decreases. This is because an increase in pressure reduces the share of recuperated heat in the reformer and an increase in steam extraction reduces the steam turbine power. Therefore, the increase in pressure does not compensate for the loss of steam turbine power due to steam extraction and the decrease of CRGT efficiency due to the reduction in recuperated heat in the reformer. Based on Fig. Fig. 5a it can be concluded that at $T = 1600$ °C the maximum CCPP efficiency of 0.6412 is at a pressure of 23 bar at $\beta = 3$.

To illustrate the effect of operational parameters on the increase in the CCPP efficiency, Fig. 5b shows the change in efficiency relative to CCPP without TCR (all steam after the steam generator is supplied to a steam turbine, $\beta = 0$). In the pressure range above 20 bar and temperature of inlet working fluid above 1400 °C (most industrially applicable level), an increase in the CCPP efficiency is about 4–7% when thermochemical exhaust heat recuperation is applied. Moreover, in the temperature range below 1200 °C the increase in the efficiency of CCPP with TCR is less than 2%. Therefore, it can be concluded that thermochemical exhaust heat recuperation has a notable effect in the high-temperature gas turbines.

The effect of operating parameters on the power of gas turbine and steam turbine cycles can be illustrated with the help of Fig. 6. Fig. 6 shows that an increase in a steam-to-methane ratio leads to decreasing in a steam turbine power and increasing in a gas turbine power. Moreover, Fig. 6 depicts that the effect of a steam-to-methane ratio at T_{in} below 1400 °C for $p = 40$ bar is negligible.

Based on data from Figs. 5 and 6, it can be concluded that the maximum efficiency of CCPPs with CRGT has in the gas turbines with an expansion ratio of 20–25, temperature of working gas of above 1500–1600 °C, and steam-to-methane ratio of 2–3. It should be noted that an increase in a steam-to-methane ratio leads to steam (water) losses because the steam for reforming is become a part of flue gases and passed to a chimney. This fact can be a reason for an increase in the operating cost of CCPP with CRGT. Therefore, a detailed techno-economic analysis should be performed after the energy analysis. A techno-economic analysis will be discussed in the future author's papers.

Energy and mass balance

A combined cycle power plant with a chemically recuperated gas turbine has a lot of energy flows in the elements of the schematic diagram. Therefore, to understand the effect of operating parameters on the efficiency of CCPP with CRGT, energy balance was compiled. Fig. 7 shows a Sankey diagram for CCPP with CRGT for the following parameters: temperature of working gas at a gas turbine inlet of 1600 °C; pressure of 20 bar (expansion/compression ratio of 20); steam-to-methane ratio of 2.0. Sankey diagram shows all notable inputs and outputs of energy flows. In this diagram, the losses in a combustion chamber, losses in the generators of a gas turbine and steam turbine, losses in a reformer and steam generator, and power in a pump are not shown.

The Sankey diagram makes it clear that the chemical energy of initial methane is transformed in a reformer into hydrogen-rich fuel. Power generated by the combustion of initial methane (100 kg/h) is 1391.6 kW. Power generated by the combustion of hot hydrogen-rich fuel after a reformer is 1890.2 kW which is 1.36 times higher than in CCPP without CRGT. Further, working gas is fed to a gas turbine to generate 749.3 kW of energy in a generator. Exhaust gas after a gas turbine is passed through a reformer to spend 303.3 kW to preheat a steam-methane mixture and supply heat for the steam methane reforming process. Exhaust gas after a reformer is supplied to a steam generator. Steam is fed to a steam turbine. In the Sankey diagram, a steam turbine is virtually divided into two parts: first steam turbine (1st. turbine) and second steam turbine (2nd ST). Steam flow after the first steam turbine is split into two streams: one for methane reforming and another for 2nd steam turbine.

Overall efficiency of the CCPP with CRGT from Fig. 7 is 0.628. The efficiency of the CCPP with a conventional gas turbine without thermochemical recuperation for the same parameters is 0.571 or 5.7% less than it is in the CCPP with CRGT. In the CCPP with CRGT 85.7% of total energy is generated in a chemically recuperated gas turbine cycle. A share of the total energy of a steam turbine cycle is 14.3%.

The Sankey diagram illustrated that optimization of a chemically recuperated gas turbine cycle is a more important task than a steam cycle because about 85% of energy is generated in a gas turbine part of a combined cycle power plant.

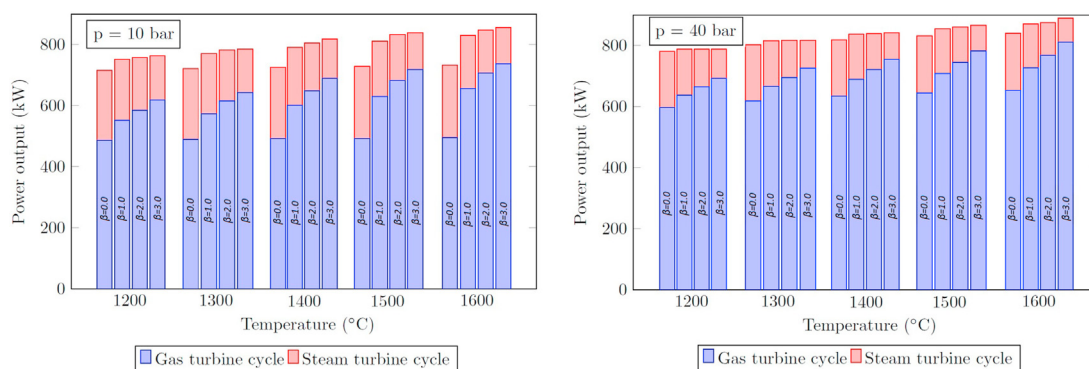


Fig. 6 – Comparison of the power of gas turbine and steam turbine cycles for various operating parameters.

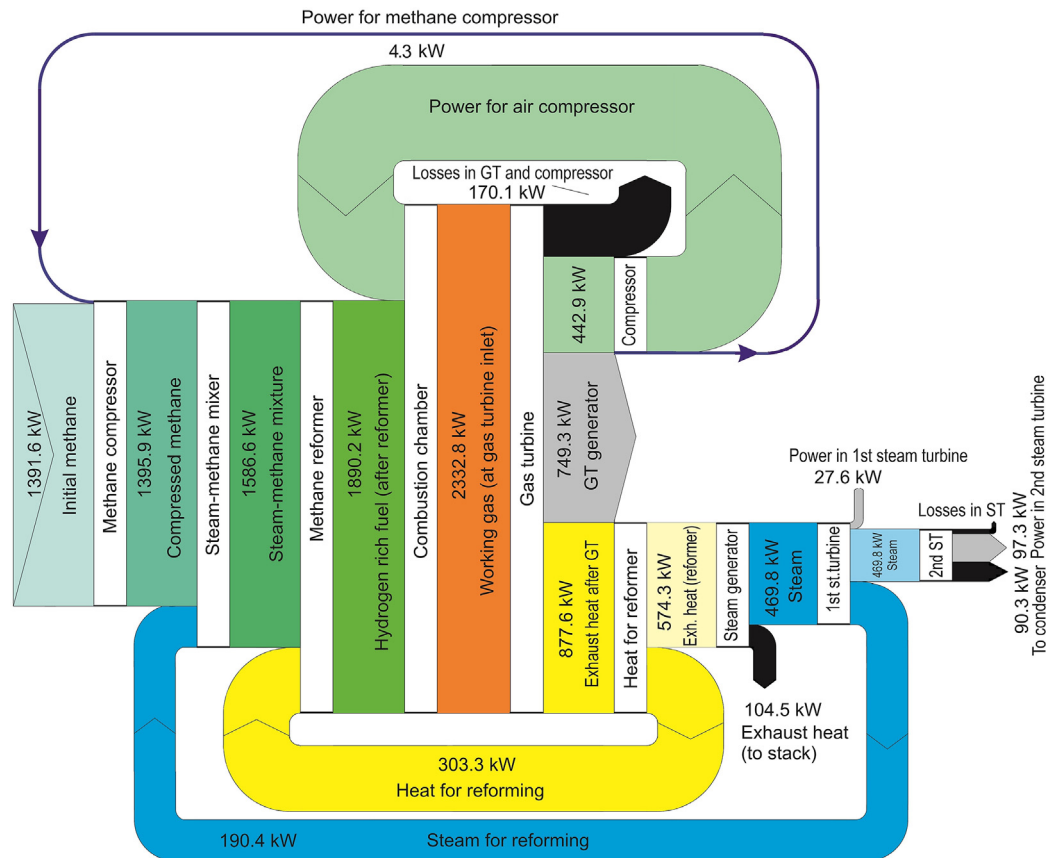


Fig. 7 – A combined cycle power plant with a chemically recuperated gas turbine: $T_{in} = 1600\text{ }^{\circ}\text{C}$, $p = 20\text{ bar}$, $\beta = 2.0$.

The combined cycle power plants with the chemically recuperated gas turbines have many ways to improve energy efficiency. One of the ways is an optimization of the parameters of the gas and steam turbine cycles. Another way is using the heat of condensation of steam from exhaust gases. An increase in a steam-to-methane ratio leads to an increase in the volume fraction of steam in exhaust gases. Increasing of volume fraction of steam in exhaust gases leads to an increase in partial pressure of steam. Therefore, condensation of steam may give an additional positive effect on energy efficiency.

The parameters of all streams in the combined cycle power plant with a chemically recuperated gas turbine for the

Sankey diagram from Fig. 7 are presented in Table 4. Table 4 shows that hydrogen-rich fuel contains a notable amount of unconverted methane. Therefore, there is a potential to increase the efficiency of CCPP with CRGT. To realize this potential, the reforming temperature should be increased. This can be at least in two cases: the first is increasing of working gas temperature (the next generation of the gas turbines will have an inlet temperature up to $1800\text{ }^{\circ}\text{C}$) and the second is increasing the exhaust gas temperature by supplementary firing. Table 4 presents that exhaust gas after a gas turbine contains about 9% of oxygen. This oxygen can be used for supplementary firing of a part of hydrogen-rich fuel. This

Table 4 – The parameters of all streams in the combined cycle power plant with chemically recuperated gas turbine for the Sankey diagram from Fig. 7.

Stream	$\dot{m}$, kg/h	T , $^{\circ}\text{C}$	p , bar	Composition, mol/mol
Initial methane (1)	100	20	10	$\text{CH}_4 = 1$
Compressed methane (2)	100	76.3	20	$\text{CH}_4 = 1$
Steam after steam generator (13)	728	400	40	$\text{H}_2\text{O} = 1$
Steam for reforming (10)	225	319.2	20	$\text{H}_2\text{O} = 1$
Steam to condenser (14)	503	36.4	0.06	$\text{H}_2\text{O} = 1$
Steam-methane mixture (3)	325	227.3	20	$\text{H}_2\text{O} = 0.67$; $\text{CH}_4 = 0.33$
Hydrogen-rich fuel (4)	325	750	20	$\text{CH}_4 = 0.13$; $\text{H}_2 = 0.43$; $\text{CO} = 0.06$; $\text{H}_2\text{O} = 0.32$; $\text{CO}_2 = 0.06$
Ambient air (15)	3472	20	1	$\text{O}_2 = 0.29$; $\text{N}_2 = 0.71$
Compressed air (9)	3472	454.8	20	$\text{O}_2 = 0.29$; $\text{N}_2 = 0.71$
Exhaust gas after GT (6)	3797	786.8	1	$\text{O}_2 = 0.09$; $\text{N}_2 = 0.68$; $\text{H}_2\text{O} = 0.18$; $\text{CO}_2 = 0.05$
Exhaust gas to chimney (8)	3797	100	1	$\text{O}_2 = 0.09$; $\text{N}_2 = 0.68$; $\text{H}_2\text{O} = 0.18$; $\text{CO}_2 = 0.05$

issue will be discussed in the author's future works. Moreover, exhaust gas after a gas turbine contains about 18% of steam. With increasing in a steam-to-methane ratio, the volume fraction of H_2O will increase. Therefore, the combined cycle power plant with a chemically recuperated gas turbine with steam condensation may be interesting for investigation. In this case, steam condensation will not only give a positive energy effect but also will compensate the make-up water for methane reforming.

Conclusion

In this paper, a combined cycle power plant with a chemically recuperated gas turbine was investigated. Thermochemical exhaust heat recuperation is employed by steam methane reforming. To understand the effect of operating parameters such as temperature, pressure, and inlet steam-to-methane ratio, the thermodynamic analysis was performed and realized via Aspen HYSYS. The results of the thermodynamic analysis show that the efficiency of CCPP with CRGT is significantly higher (4–7%) than the efficiency of CCPP with a conventional gas turbine without TCR. Maximum efficiency of CCPP with CRGT of 0.6412 is observed at $T = 1600\text{ }^{\circ}\text{C}$ at pressure of 23 bar for a steam-to-methane ratio of 3.0. In the temperature range of inlet working gas below $1200\text{ }^{\circ}\text{C}$ the increase in the efficiency of CCPP with TCR is less than 2%. Therefore, it can be concluded that thermochemical exhaust heat recuperation has a notable effect for the high-temperature gas turbines.

A Sankey diagram was compiled to show the energy flows in CCPP with CRGT at $T_{in} = 1600\text{ }^{\circ}\text{C}$, $p = 20\text{ bar}$, $\beta = 2.0$. The efficiency of CCPP with CRGT at these parameters is 0.628. The efficiency of the CCPP with a conventional gas turbine without thermochemical recuperation for the same parameters is 0.571 or 5.7% less than it is in the CCPP with CRGT. In the CCPP with CRGT 85.7% of total energy is generated in a chemically recuperated gas turbine cycle. A share of the total energy generated in a steam turbine cycle is 14.3%. The Sankey diagram illustrates that optimization of a chemically recuperated gas turbine cycle is a more important task than a steam cycle because about 85% of energy is generated in a gas turbine part of a combined cycle power plant.

An analysis of mass balance showed that exhaust gas after a gas turbine has about 18% of H_2O and 9% of O_2 (for $\beta = 2.0$, $p = 20\text{ bar}$, $T_{in} = 1500\text{ }^{\circ}\text{C}$) as well as hydrogen-rich fuel contains about 13% of unreacted methane. Therefore, there is the potential to improve the efficiency of CCPP with CRGT in various ways, for example, supplementary firing to increase the temperature for methane reforming; condensation of H_2O to get a high heating value.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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